

Here's a critical (but friendly) review of the uploaded DeXposure-FM draft, with concrete improvement suggestions.

1) Framing + contribution clarity

- The paper's core claim is *foundation-model-style transfer* to DeFi exposure forecasting, but the current text reads more like "fine-tune a pretrained encoder + add heads." You *do* state GraphPFN is the "core encoder" and explain frozen vs fine-tuned modes DeXposure_FM — but to sell "foundation model," you should sharpen what is *general-purpose* about DeXposure-FM (e.g., pretraining objectives, breadth of tasks, transfer to unseen chains/protocol types, scaling laws, or emergent capabilities).
- Right now the "foundation" aspect leans heavily on GraphPFN being pretrained elsewhere DeXposure_FM. If the novelty is "foundation model on DeXposure," readers will ask: what is pretrained on what, exactly, and what is the new pretraining objective vs just downstream finetuning?

Suggestion: Add a short "What makes this a foundation model?" paragraph early, with 3–4 crisp criteria (multi-task + reusable representations + transfer across regimes/entities + scalable training), and map your design to each.

2) Experimental completeness + internal consistency (high priority)

- Table 3 has "TBD" entries for the important long horizon $h = 14$ for DeXposure-FM DeXposure_FM, and later the text explicitly says $h = 14$ results are pending due to compute DeXposure_FM. That's honest, but it weakens the paper's "forecasting stability" claim—especially because long-horizon performance is where "foundation" representations should matter most.
- There's also a factual inconsistency: the "Validity and Scope" section says ROLAND has AUROC 0.718 at $h = 14$ DeXposure_FM, but Table 3 reports ROLAND AUROC 0.955 at $h = 14$ DeXposure_FM. That kind of mismatch will cause referee bloodlust.

Suggestions:

- Finish the missing $h = 14$ runs or remove the horizon from the main claims until it's done.
- Audit every numeric claim against tables (especially anything used to make an argument).
- Add uncertainty bands or at least bootstrap standard errors for key metrics (AUROC/AUPRC/MAE), because the dataset is huge and "everything is significant" unless you're careful.

3) Evaluation design: beware "persistence wins" and leakage-by-structure

Your dataset has **98.5% mean edge overlap week-to-week** DeXposure_FM. You correctly note that high recall in imputation "directly reflects" that overlap DeXposure_FM. That's good scientific hygiene—but it also means many forecasting metrics may be inflated by a trivial strategy: "predict last week's graph."

Suggestions (strong):

- Add naive persistence baselines explicitly:
 - Edge existence: $\hat{A}_{t+h} = A_t$
 - Edge weight: $\hat{W}_{t+h} = W_t$ (or last observation carried forward on log-scale)
 - Node TVL: random walk / AR(1) on $\log(1 + \text{TVL})$
- Report **incremental skill** over persistence (e.g., ΔAUPRC , relative MAE reduction) and evaluate performance separately on *edge changes* (new edges, deleted edges, large weight

jumps). This will show whether the model predicts *dynamics*, not just inertia.

4) Negative sampling and edge prediction realism

You sample negative edges uniformly from the complement with a 5:1 ratio DeXposure_FM. Uniform negatives are easy: most non-edges are “obviously” non-edges (far in sector space, never share tokens, etc.). That can artificially boost AUROC/AUPRC.

Suggestions:

- Use **hard negatives**: sample non-edges that are plausible (same sector pairs, share token universe, or historically existed but disappeared).
- Report results under multiple negative-sampling schemes, or evaluate link prediction as ranking over a candidate set that mimics a real monitoring use-case.

5) Missingness / imputation task: MCAR is too kind

Your imputation masks edges under MCAR (Missing Completely at Random) DeXposure_FM. In practice, missingness is usually MNAR-ish (Missing Not At Random): APIs fail more on smaller chains, noisy protocols, or during congestion and crises.

Suggestions:

- Add structured missingness: mask entire chains, sectors, or low-liquidity protocols; mask more during stress windows; mask edges with small weights (measurement noise) vs large weights (critical exposures).
- Evaluate downstream impact: does imputation improve later forecasting/link prediction? (You already allude to practical uses DeXposure_FM —make it quantitative.)

6) Stress/scenario analysis: good direction, but the contagion model is very stylized

You acknowledge the contagion model assumes linear loss propagation and omits feedback (liquidation cascades, etc.) DeXposure_FM, and you also note that weekly aggregation smooths out flash loans and rapid liquidations DeXposure_FM. Great—this is exactly where critics will poke.

Suggestions:

- Position the contagion exercise as a **first-order structural stress map**, not a full DeFi microstructure simulator.
- Add sensitivity checks: compare outcomes across alternative propagation rules or shock sizes; show whether “bridges dominate” is robust (you cite a big bridge loss example DeXposure_FM).
- If possible, incorporate at least one *feedback proxy*: e.g., liquidation threshold triggers or a simple price-impact function, even if stylized.

7) Systemic Importance Score and policy outputs: define, justify, and validate

You describe SIS as combining PageRank, TailExposure, and TVL DeXposure_FM. That’s plausible, but it will be attacked as ad hoc unless you provide:

- explicit functional form and weights,
- sensitivity analysis,

- and some external validation (e.g., does SIS spike for protocols that later suffered distress?).

Suggestions:

- Treat SIS as a **model-based index**: calibrate weights by predicting historical “bad outcomes” (exploits, depegs, large drawdowns) or by matching known systemic events.
- Provide rank stability over time and under perturbations (e.g., token price noise), especially since you caution TVL is imperfect `DeXposure_FM`.

8) Paper polish: placeholders and “draft smell”

There are still placeholders like “Section ??” `DeXposure_FM`, and partially completed results tables `DeXposure_FM`. Reviewers are allergic to this.

Suggestion: Before circulating widely, do a “no placeholders” sweep + consistency checks (definitions, symbols, and references). It’s boring work, but it prevents the dumbest possible rejection.